

StormSense

AI-Driven Hyper-Local Severe Weather Early Warning & Nowcasting System

West Bengal, India

Project Summary Document

Generated from live inspection of the running repository and application

1. What Is StormSense?

StormSense is a short-range severe weather warning system for West Bengal, India. It looks at the current state of the atmosphere and predicts, for the next 2 to 6 hours, how likely severe thunderstorms are, how much rain is expected, and where flash-flooding is more likely, on a fine 0.25-degree grid (roughly 25-28 km per cell).

Nowcasting means very short-range forecasting -- not tomorrow's weather, but the next few hours. This matters because thunderstorms and heavy rain cells can form and move within an hour or two, faster than a normal daily forecast can react to.

- Who could use it: disaster management cells, local administrations, and the public in West Bengal wanting a short-range, location-aware severe-weather signal.
- What makes it useful: it combines a real trained deep-learning model with live NOAA GFS atmospheric data, and keeps a completely separate, verifiable historical case study (Cyclone Remal, 2024) to demonstrate the pipeline on a known severe event.

2. The Problem We Are Solving

Severe convective weather -- thunderstorms, sudden heavy rain, squalls -- is hard to forecast at a local scale because it develops fast and is smaller than the grid of a typical daily national forecast. A city-level or state-level forecast can say rain is expected today without telling a specific district whether it will see a dangerous storm in the next two hours.

- Example: a Kalbaishakhi-type pre-monsoon squall or a landfalling cyclone like Remal can bring serious thunderstorm and flash flood risk to one district while a neighbouring district stays dry.
- Ordinary forecasts do not answer: what is my flash-flood risk in the next 2 hours, in my district, right now? StormSense targets exactly that gap.

3. Our Solution (Plain English)

Input weather data -> Data processing -> AI model -> Risk prediction -> Maps / district warnings / dashboard

In practice: StormSense pulls a real snapshot of the atmosphere (from NOAA's GFS model in live mode, or from a frozen 2024 ERA5 record for the historical case study), reshapes it into the exact format its neural network expects, runs the network, and turns the raw numbers into a colour-coded risk map, per-district cards, and a dashboard.

4. Complete System Architecture

[A] Verified directly from the code (src/inference/, weather-app/backend/main.py).

LIVE MODE:

NOAA GFS f000 analyses (AWS Open Data S3)

- > `src/inference/gfs_live.py` (fetch + harmonize into ERA5-shaped tensors)
- > `src/inference/nowcast_service.py` (`NowcastService.refresh_live_state`)
- > `src/inference/predictor.py` (`NowcastPredictor.predict` -> `SevereWeatherNetV2`)
- > `risk_map.py` / `risk_surface.py` / `xai.py` (turn model output into GeoJSON, PNGs, factor explanations)
- > `weather-app/backend/main.py` (FastAPI, 44 route handlers)
- > `weather-app/index.html` + `js/app.js` (Leaflet map + dashboard, vanilla JS, no build step)

HISTORICAL MODE:

ERA5 reanalysis memmap cache (`processed/cache/era5_memmap`)

- > `NowcastService._init_operational_state()` locates the exact Cyclone Remal analysis window (2024-05-26T12:00:00Z)
- > same predictor / risk / XAI / frontend pipeline as live mode, but reading a frozen array instead of a live feed.

Important: Live and Historical are two separate state objects inside `NowcastService` (`live_pred` vs `current_pred`) and are never merged. [A] Verified in `src/inference/nowcast_service.py`.

5. Data Sources

Source	Provides	Live/Historical	Enters Pipeline At	Category
NOAA GFS 0.25° f000 analysis (AWS S3)	Real 0-hour atmospheric analysis: wind, temp, dewpoint, pressure, CAPE, CIN, humidity, precipitation rate	Live	<code>src/inference/gfs_live.py</code>	Observation / Model Input
ERA5 reanalysis (frozen cache)	Same variable set, historical record used only for the Cyclone Remal case study (2024-05-26 12:00 UTC)	Historical	<code>src/inference/nowcast_service.py</code>	Historical Archive / Model Input
SRTM Digital Elevation Model	Static terrain elevation, 0.25° resampled	Static	Model input + flash-flood terrain factor	Static Input
OpenWeatherMap current weather (24 WB stations)	Real current station observations	Live	<code>observation_surface.py</code>	Observation

RainViewer public radar API	Composite weather radar tiles	Live (~10 min frames)	Frontend, Radar & Satellite Feeds view	Observation
INSAT-3DR (ISRO/SAC) HDF5 archive	Episodic satellite channels: precip, cloud-top pressure, humidity, cloud mask	Historical archive	src/inference/insat_archive.py	Historical Archive
v2_calibrated_best.pt	Trained SevereWeatherNetV2 weights + calibration	N/A	src/inference/predictor.py	Model Artifact

Note: RainViewer is observed radar reflectivity, not a rainfall amount, and it is NOT shown on the historical Cyclone Remal case study because no verified Remal radar archive exists in this project. [A]

6. The AI Model

Model: SevereWeatherNetV2, a tri-stream ConvGRU network, approximately 781,889 trainable parameters. [A] src/models/v2_model.py

ConvGRU in simple terms: a normal GRU is good at remembering patterns over time. A Convolutional GRU does the same, but for a 2D grid of weather values -- so it remembers how a pattern is moving and changing across the map, hour by hour, for the 6 hours leading up to now.

- Input: 6 consecutive hourly snapshots of the atmosphere (t-5h ... t0)
- Streams: surface variables, pressure-level wind, pressure-level thermodynamics -- separate ConvGRUs combined later
- Output: 5 forecast horizons at once (+2h, +3h, +4h, +5h, +6h)
- Multi-task: predicts BOTH severe-weather probability AND 3-hour rainfall
- Calibration: temperature scaling + a per-lead-time probability threshold, from the checkpoint metadata

We do NOT claim: a specific accuracy percentage as a headline number. Measured skill is reported honestly in Section 24 and Document 3.

7. What Goes Into the Model?

Variable	Meaning	Why It Matters	Source
u10, v10	10m wind (E-W, N-S)	Wind speed/direction; shear and convergence	GFS/ERA5
d2m	2m dewpoint	Moisture near the	GFS/ERA5

	temperature	surface	
t2m	2m air temperature	Surface heating -- drives instability	GFS/ERA5
sp	Surface pressure	Basic state variable	GFS/ERA5
CAPE	Convective Available Potential Energy -- fuel for rising, storm-forming air	Core instability signal	GFS/ERA5
CIN	Convective Inhibition -- a lid suppressing storm formation	Low CIN + high CAPE = storms trigger easily	GFS/ERA5
TCWV	Total Column Water Vapour	More moisture = more potential rainfall	GFS(PWAT)/ERA5
TP	Total precipitation rate	Direct precipitation signal	GFS(PRATE)/ERA5
u,v,z,q,t (pressure levels)	Wind, height, humidity, temperature at 250-1000 hPa	Vertical structure -- shear, moisture layering	GFS/ERA5
DEM	Static ground elevation	Terrain forcing, flash-flood terrain factor	SRTM

Load-bearing detail: Pressure levels are stored ASCENDING in code: [250, 300, 500, 700, 850, 1000]. The config YAML lists them descending for human reading only -- the model slices by index, so this must never be reordered. [A]

8. What Comes Out of the Model?

- Severe-weather probability (0-100%): calibrated estimate of severe convective weather at that grid cell and horizon
- 3-hour rainfall prediction (mm): a regression output
- Flash-flood risk PROXY (%): NOT a direct model output. Calculated afterward from rainfall + severe probability + terrain steepness. Explicitly a proxy, never an official flood forecast

Important: Proxy means an indirect stand-in calculation, not a directly measured quantity. A probability means a calibrated likelihood estimate, not a certainty.

9. NOW / +2h / +4h / +6h

Issue time = the exact instant a forecast is generated for. Valid time = issue time + lead hours. Always shown separately and computed to the exact second.

- NOW: the current observed state -- not a model forecast
- +2h/+4h/+6h: the model's forecast for issue_time + 2/4/6 hours

Example: if the live analysis is issued at 14:00 UTC, +2h means the forecast valid at 16:00 UTC.

10. Live Mode

When Live mode is open: NOW shows real current station observations (OpenWeatherMap, 24 stations) interpolated into a blue/teal map layer, plus real live radar (RainViewer) on the Radar & Satellite Feeds page. +2h/+4h/+6h switch the map to the model's forecast risk surface, generated from the latest GFS f000 analysis (typically several hours old due to NOAA publication latency -- disclosed, not hidden).

Separation kept strict: current observation, model input, and model output are three different things in the code and UI, never relabelled as each other. [A]

11. Historical Mode -- Cyclone Remal

Cyclone Remal was a real Severe Cyclonic Storm that made landfall on the West Bengal/Bangladesh coast on 26 May 2024. StormSense uses the real ERA5 reanalysis state at 2024-05-26T12:00:00Z as a frozen, repeatable case study.

- Historical NOW (T0): the real ERA5 analysis at 12:00 UTC, shown as observed rainfall rate (not a forecast)
- +2h/+4h/+6h: the model's genuine forecast run from that t0 state, valid at 14:00/16:00/18:00 UTC
- Historical radar: genuinely unavailable -- no verified archived Remal radar exists, and the UI states this plainly
- Historical satellite (INSAT-3DR): real ISRO archive files exist, but nearest acquisitions are 21-35 hours from the case-study window, so reported as unavailable for this specific target

Strict separation: Historical mode never uses the current wall-clock, current GFS analysis, or live station data, and vice versa. [A]

12. Radar

Radar measures dBZ (decibels of reflectivity) -- how much energy bounces back off raindrops/hail. Higher dBZ generally means heavier precipitation. RainViewer is a real, live, composite radar mosaic, refreshed roughly every 10 minutes, shown ONLY on Radar & Satellite Feeds and ONLY in Live mode.

Critical distinction: The AI forecast risk heatmap is NOT radar. It is the model's own probability output on a 4-band colour scale. Visually similar, meteorologically different.

13. Satellite

INSAT-3DR is ISRO's geostationary meteorological satellite. This project holds a real, on-disk, episodic archive of Level-2B products (verified readable HDF5 files with genuine ISRO/SAC timestamps).

What it is NOT: Not a live satellite feed. Historical, episodic, used only for qualitative corroboration, and no imagery is close enough in time to Remal's analysis window to display as contemporaneous. [A]

14. District Advisories

Each of the 23 West Bengal districts maps to the grid cells inside (or within 0.15 degrees of) its polygon. District risk numbers are the MAXIMUM and 90th-percentile values across those cells -- a civil-protection-style aggregation, not a simple average. [A]

15. XAI Feature Attribution

Honest framing: Rule-based, physics-inspired attribution. NOT SHAP, NOT LIME, not gradient-based, does not inspect network weights. [A] `src/inference/xai.py`

Factor	What it means	How Scored	Limitation
Rainfall	Recent 1h/3h/6h rainfall	Weighted sum, heaviest on most recent hour	Fixed formula, not learned
Moisture/Humidity	RH + dewpoint	Rises above 60% RH, boosted above 24-26C dewpoint	Threshold-based
Wind	10m wind speed	Linear scale to 30 km/h	No direction/shear structure
Instability (CAPE)	Convective available potential energy	Piecewise scale	Physical proxy, not internal network reasoning

16. GIS

The map is a 33x25 grid (0.25 degree) over 20-28N, 84-90E, Leaflet + Esri dark basemap. District boundaries from a verified 23-district GeoJSON. Clicking a point queries `/api/nowcast/point` for the nearest grid cell and shows that cell's forecast for the current mode and horizon.

17. Thresholds

Band	Range	Meaning
Normal	<25%	Low modeled severe-weather probability

Watch	25-50%	Elevated -- monitor
Alert	50-75%	High modeled risk
Warning	≥75%	Very high modeled risk

Important: Internal UI bands from the model's calibrated probability. NOT an official IMD warning category.

18. Streams / Data Feeds

Shows operational status of each ingestion pipeline (GFS fetch, station observations, model inference) -- demonstrates a monitored live pipeline, not a static demo.

19. Frontend

Vanilla JavaScript (no React, no build step, no npm), served as static files by FastAPI, Leaflet for maps, Tailwind CSS via CDN. [A] No package.json or node_modules exist in weather-app/.

20. Backend

FastAPI (Python), 44 route handlers, serving both the JSON API and static frontend files. Inference runs in-process via a cached singleton NowcastService.

21. Complete Request Flow

User opens dashboard -> Frontend calls /api/nowcast/summary -> Backend resolves mode -> Reads cached prediction (or triggers a fresh GFS fetch) -> Aggregates risk per district/cell -> Returns JSON -> Frontend paints map, KPI cards, forecast strip -> User sees the dashboard.

22. What Makes This a Hackathon Project?

The engineering contribution: (1) a real multi-task ConvGRU trained on ERA5 predicting both probability and rainfall at 5 lead times, (2) a live GFS pipeline that structurally prevents feeding future forecast hours as if they were the past (a real leakage bug found and fixed during development), and (3) strict, code-enforced separation between live, historical, observed, and forecast data.

23. What The Project Can And Cannot Claim

We CAN claim:

- A real trained deep-learning model producing calibrated severe-weather probabilities and rainfall forecasts
- A genuine live NOAA GFS ingestion pipeline with disclosed analysis latency
- A fully separated, reproducible historical case study on a real cyclone (Remal)

- Honest disclosure when a data product is unavailable, rather than fabrication

We should NOT claim:

- 100% accurate or any unqualified accuracy figure
- Official government warning
- Real radar for the AI heatmap
- True explainable AI for the rule-based XAI layer
- Live satellite for the historical INSAT archive

24. Limitations

- Live forecast skill is measurably weaker than ERA5 held-out test metrics due to GFS/ERA5 distribution shift
- False-alarm rate (FAR) on held-out ERA5 test data is roughly 0.59 mean across leads
- GFS analysis latency means the live current atmospheric input is typically several hours old
- No archived Remal radar or contemporaneous INSAT imagery exists in this project
- XAI is a physics-inspired heuristic, not a validated explanation of the network's internal reasoning

25. 2-Minute Explanation

StormSense is a severe-weather early-warning system for West Bengal. It takes a live snapshot of the atmosphere from NOAA's GFS model, feeds six hours of that data into a trained ConvGRU deep-learning model, and produces a 0-6 hour forecast of severe-weather probability and rainfall on a fine grid across the state. Those numbers become a colour-coded risk map, per-district advisory cards, and plain dashboard KPIs. We also built a fully separate historical case study on Cyclone Remal (May 2024) using real ERA5 reanalysis data, so the same pipeline can be demonstrated end-to-end on a known severe event. We were strict about not faking data: live and historical never mix, radar is never confused with the AI forecast, and when a data product genuinely isn't available -- like archived Remal radar -- we say so instead of inventing it.

26. 30-Second Pitch

StormSense predicts severe thunderstorms, rainfall, and flash-flood risk for West Bengal, 2 to 6 hours ahead, using a real deep-learning model fed by live NOAA weather data -- and we can demonstrate it live on both current conditions and a real historical cyclone.

27. Terms Cheat Sheet

Term	One-line meaning
Nowcasting	Very short-range forecasting (0-6 hours)
CAPE	Fuel available for thunderstorms to form

CIN	A lid that suppresses storms from forming
TCWV	Total moisture in the air column above a point
dBZ	Radar reflectivity unit -- higher usually means heavier precipitation
ConvGRU	A neural network that remembers pattern changes over time AND space
Proxy	An indirect calculated stand-in, not a direct measurement
Issue time	The instant a forecast is generated for
Valid time	The future instant a forecast describes
Calibration	Adjusting raw model scores so a probability means what it says